4.00

ت (٩)

(L'HOPital's Rule)

النهايات وقاعدة لوبيتال

+ حل شيت الدرس من الفرقة الأولى

L'Hopital's Rule ت (9)

(قاعدة لوبيتال)

$$\lim_{x \to a} \frac{f(x)}{g(x)} = \frac{f(a)}{g(a)} = \boxed{}$$

Determined Value
(كمية معينة)

$\Downarrow$

Any finite value

Undetermined Value
(كمية غير معينة)

حفظ

$\Downarrow$

$\left(\frac{\infty}{\infty}\right)$, $\left(\frac{0}{0}\right)$

$(0 * \infty)$

$(\infty - \infty)$

$(1)^{\infty}$, $(\infty)^{0}$, $(0)^{0}$

← في حالة أن ناتج Lim كمية معينة تنتهي المسألة

← في حالة أن ناتج Lim كمية غير معينة نستخدم قاعدة L'hopital للتحويل إلى كمية معينة

* L'Hopital's rule :-

$$\text{If} \quad \lim_{x \to a} \frac{f(x)}{g(x)} = \begin{cases} \left(\frac{0}{0}\right) \\ \text{or} \\ \left(\frac{\infty}{\infty}\right) \end{cases}$$

$$= \lim_{x \to a} \frac{f'(x)}{g'(x)} \Rightarrow$$ (if the result $\frac{0}{0}$ or $\frac{\infty}{\infty}$ we apply L'hopital again)

<u>ملاحظة</u> (لا يمكن تطبيق قاعدة L'Hopital إلا إذا كان الناتج $\frac{0}{0}$ أو $\frac{\infty}{\infty}$ فقط)

في حالة أن الناتج كمية غير معينة غير $\frac{0}{0}$ أو $\frac{\infty}{\infty}$ نحول أولاً إلى $\frac{0}{0}$ أو $\frac{\infty}{\infty}$ ثم نستخدم L'hopital

→ $(0 * \infty)$: نقلب أحد الدالتين في المقام

→ $(\infty - \infty)$: توحيد مقامات
(أو نضرب في المرافق لو مفيش مقام)

→ $(1)^{\infty}, (\infty)^{0}, (0)^{0}$: نأخذ ln للطرفين

* Notes:

* $\frac{1}{0} = \infty$ * $\frac{1}{\infty} = 0$ * $\infty \cdot \infty = \infty$

* $\frac{\infty}{0} = \infty$ * $\frac{0}{\infty} = 0$ * $\infty + \infty = \infty$

* $e^{\infty} = \infty$ * $e^{-\infty} = 0$ * $e^{0} = 1$

* $\ln 1 = 0$ * $\ln \infty = \infty$ * $\ln 0 = -\infty$

* $(0)^{\infty} = 0$ * $(\infty)^{\infty} = \infty$ * $\tan^{-1} \infty = \frac{\pi}{2}$

* $\cot^{-1} \infty = 0$ * $\cot 0 = \infty$ * $\tan \frac{\pi}{2} = \infty$

* (رقم أكبر من 1)$^{\infty} = \infty$ * (رقم أكبر من 1)$^{-\infty} = 0$ * (رقم أقل من 1)$^{\infty} = 0$ * (رقم أقل من 1)$^{-\infty} = \infty$

← صور نهايات مشهورة محفوظة:

* $\lim_{x \to 0} \frac{\sin ax}{x} = a$

* $\lim_{x \to 0} \frac{\tan ax}{x} = a$

* $\lim_{x \to \infty} \frac{\sin a/x}{1/x} = a$

* $\lim_{x \to \infty} \frac{\tan a/x}{1/x} = a$

تابع صور نهايات مشهورة محفوظة

$$\lim_{x \to \infty} \frac{\text{Polynomial of degree } (m)}{\text{Polynomial of degree } (n)}$$

أكبر أس في الدالة (m)

أكبر أس (n)

If $\boxed{m > n}$ — $\lim = \infty$

If $\boxed{m < n}$ — $\lim = \text{zero}$

If $\boxed{m = n}$ — $\lim = \dfrac{\text{معامل أكبر أس}}{\text{معامل أكبر أس}}$

Ex: $\displaystyle \lim_{x \to \infty} \frac{3x^2 - 4x + 1}{x + 7} = \infty$

Ex: $\displaystyle \lim_{x \to \infty} \frac{2x^3 + 4x - 1}{x^5 + x^2 - 4} = 0$

Ex: $\displaystyle \lim_{x \to \infty} \frac{3x^2 + x - 1}{2x^2 - 4x + 7} = \frac{3}{2}$

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note If $\lim = \pm\infty$

$\therefore$ we say Limit $\boxed{\text{does not exist}}$

لو الناتج في أي مسألة طلع $\pm\infty$ نسمي Lim إنه

غير موجود

Ex ① $\lim_{x\to\infty} \dfrac{x^2}{e^{x^2}} = \dfrac{\infty}{\infty}$

using L'hopital

$= \lim_{x\to\infty} \dfrac{\cancel{2x}}{\cancel{2x}\, e^{x^2}} = \dfrac{1}{\infty} = \boxed{0}$

Ex ② $\lim_{x\to 1} \left(\dfrac{x}{x-1} - \dfrac{1}{\ln x} \right) = (\infty - \infty)$

بتوحيد المقامات

$= \lim_{x\to 1} \dfrac{x \ln x - x + 1}{(x-1)\ln x} = \dfrac{0}{0}$

using L'hopital

$= \lim_{x\to 1} \dfrac{\cancel{x \cdot \frac{1}{x}} + \ln x \cancel{- 1}}{(x-1)\cdot\frac{1}{x} + \ln x}$

$= \lim_{x\to 1} \dfrac{\ln x}{1 - \frac{1}{x} + \ln x} = \dfrac{0}{0}$

using L'hopital again

$= \lim_{x\to 1} \dfrac{1/x}{\frac{1}{x^2} + \frac{1}{x}} = \boxed{\dfrac{1}{2}}$

Ex ③ $\lim\limits_{x \to \frac{\pi}{4}} \dfrac{\sec^2 x - 2\tan x}{1 + \cos 4x} = \dfrac{0}{0}$

Using L'hopital

$\sec\frac{\pi}{4} = \sqrt{2}$

$\tan\frac{\pi}{4} = 1$

$\cos\pi = -1$

$\sin\pi = 0$

$$= \lim_{x \to \frac{\pi}{4}} \frac{2\sec^2 x \tan x - 2\sec^2 x}{-4\sin 4x} = \frac{0}{0}$$

قبل تطبيق L'hopital مرة أخرى نحاول
تبسيط المسألة أو أخذ عامل مشترك
ونعوض فيه بشرط أن يكون الناتج أي رقم
عدا 0 أو ∞.

$$= \lim_{x \to \frac{\pi}{4}} \underbrace{\frac{2\sec^2 x}{-4}}_{-1} \cdot \frac{\tan x - 1}{\sin 4x}$$

$$= -\lim_{x \to \frac{\pi}{4}} \frac{\tan x - 1}{\sin 4x} = \frac{0}{0}$$

$$= -\lim_{x \to \frac{\pi}{4}} \frac{\overset{2}{\sec^2 x}}{4\underset{-1}{\cos 4x}} = \boxed{\frac{1}{2}}$$

Ex (4) $\lim_{x\to 0} (x)^{\tan x} = (0)^{\infty}$

let $y = \lim_{x\to 0} (x)^{\tan x}$

بأخذ ln للطرفين

$\therefore \ln y = \lim_{x\to 0} \tan x \cdot \ln x = 0 \cdot -\infty$

لتطبيق أحد الاشكال المعتاد

$= \lim_{x\to 0} \frac{\ln x}{\cot x} = \frac{-\infty}{\infty}$

using L'hopital

$= \lim_{x\to 0} \frac{1/x}{-\csc^2 x}$

$= \lim_{x\to 0} -\frac{\sin^2 x}{x} = \lim_{x\to 0} (-\underbrace{\sin x}_{0}) \cdot \underbrace{\frac{\sin x}{x}}_{1}$

$= 0$

$\therefore \ln y = 0$

$\therefore y = e^0 = \boxed{1}$

Ex ⑤ $\lim\limits_{x \to 0} \dfrac{e^{x} + \sin x - 1}{\ln(1+x)} = \dfrac{0}{0}$

using L'hopital

$$= \lim_{x \to 0} \frac{e^{x} + \cos x}{\frac{1}{1+x}} = \frac{1+1}{1} = \boxed{2}$$

Ex ⑥ $\lim\limits_{x \to \frac{\pi}{2}} (\tan x)^{\cos x} = (\infty)^{0}$

let $y = \lim\limits_{x \to \frac{\pi}{2}} (\tan x)^{\cos x}$

$$\therefore \ln y = \lim_{x \to \frac{\pi}{2}} \cos x \cdot \ln \tan x = 0 \cdot \infty$$

بقلب أحد الاثنين

$$= \lim_{x \to \frac{\pi}{2}} \frac{\ln \tan x}{\sec x} = \frac{\infty}{\infty}$$

using L'hopital

$$= \lim_{x \to \frac{\pi}{2}} \frac{\frac{1}{\tan x} \cdot \sec^2 x}{\cancel{\sec x} \tan x} = \lim_{x \to \frac{\pi}{2}} \frac{\sec x}{\tan^2 x}$$

$$= \lim_{x \to \frac{\pi}{2}} \frac{\frac{1}{\cos x}}{\frac{\sin^2 x}{\cos^2 x}} = \lim_{x \to \frac{\pi}{2}} \frac{\cos x}{\sin^2 x} = 0$$

$$\therefore \boxed{y = e^{0} = 1}$$

Ex ⑦: $\lim_{x \to \infty} \left(\cos \frac{2}{x}\right)^{x^3} = (1)^{\infty}$

let $y = \lim_{x \to \infty} \left(\cos \frac{2}{x}\right)^{x^3}$

بأخذ ln للطرفين

$\ln y = \lim_{x \to \infty} x^3 \cdot \ln \cos \frac{2}{x} = \infty * 0$

$= \lim_{x \to \infty} \frac{\ln \cos \frac{2}{x}}{\left(\frac{1}{x^3}\right)} = \frac{0}{0}$

Using L'Hopital

$= \lim_{x \to \infty} \frac{\left(\frac{\cancel{-} \sin \frac{2}{x}}{\cos \frac{2}{x}} \cdot \frac{\cancel{-}2}{\cancel{x^2}}\right)}{\frac{-3}{\cancel{x^4}\, x^2}}$

$= \lim_{x \to \infty} \frac{\sin \frac{2}{x}}{\left(\frac{1}{x}\right)} \cdot \frac{\left(\frac{2}{\cos \frac{2}{x}}\right)}{\frac{-3}{x}}$

②

$= \lim_{x \to \infty} \frac{4}{-3} \cdot \frac{x}{\cos \frac{2}{x}} = \frac{4}{-3} \cdot \frac{\infty}{1} = -\infty$

$\therefore \ln y = -\infty$

$\therefore y = e^{-\infty} = \boxed{0}$

Alexandria University
Faculty of Engineering
Civil Engineering Department

Mathematics 3
February 2017
First Year

Sheet (2): Indeterminate forms and L'Hôpital's rule

1- $\lim\limits_{x\to-\infty} \dfrac{x+1}{x^2+4x+3}$

2- $\lim\limits_{x\to 0} \dfrac{e^{2x}-1}{x}$

3- $\lim\limits_{x\to 0} \dfrac{\sin x}{e^{3x}-1}$

4- $\lim\limits_{x\to 0} \dfrac{\sin x - x}{x^3}$

5- $\lim\limits_{x\to\infty} \dfrac{x^3}{e^x}$

6- $\lim\limits_{x\to\infty} \dfrac{e^x}{x^3}$

7- $\lim\limits_{x\to 0} \dfrac{x\cos x - \sin x}{x\sin^2 x}$

8- $\lim\limits_{x\to 1} \dfrac{\sin \pi x}{x-1}$

9- $\lim\limits_{x\to\infty} \dfrac{\ln x}{\sqrt{x}}$

10- $\lim\limits_{x\to 1} \dfrac{\ln(\ln x)}{\ln x}$

11- $\lim\limits_{x\to 0} \dfrac{\sin(\sin x)}{\sin x}$

12- $\lim\limits_{x\to 0^+} \dfrac{\sin x}{\sqrt{x}}$

13- $\lim\limits_{x\to\infty} (\sqrt{x^2+1} - x)$

14- $\lim\limits_{x\to 0}\left(\dfrac{1}{\sqrt{x}} - \sqrt{\dfrac{x}{x+1}}\right)$

15- $\lim\limits_{x\to\infty} x\tan\left(\dfrac{1}{x}\right)$

16- $\lim\limits_{x\to 0} x^2 \ln x$

17- $\lim\limits_{x\to 0} (1-2x)^{1/x}$

18- $\lim\limits_{x\to 0} \dfrac{5^x - 3^x}{x}$

19- $\lim\limits_{x\to\infty} \dfrac{3^x}{5^x}$

20- $\lim\limits_{x\to 0} \dfrac{\cos x}{x^2}$

21- $\lim\limits_{x\to\infty} \left(1+\dfrac{1}{x}\right)^x$

22- $\lim\limits_{x\to 0} x\cot 2x$

23- $\lim\limits_{x\to 0} (\cos x)^{1/x}$

Sheet(2): L'Hopital's Rule

(1) $\lim\limits_{x\to\infty} \dfrac{x+1}{x^2+4x+3} = \dfrac{\infty}{\infty}$

using L'Hopital

$= \lim\limits_{x\to\infty} \dfrac{1}{2x+4} = \dfrac{1}{\infty} = \boxed{0}$

(2) $\lim\limits_{x\to 0} \dfrac{e^{2x}-1}{x} = \dfrac{1-1}{0} = \dfrac{0}{0}$

$= \lim\limits_{x\to 0} \dfrac{2e^{2x}}{1} = \boxed{2}$

(3) $\lim\limits_{x\to 0} \dfrac{\sin x}{e^{3x}-1} = \dfrac{0}{1-1} = \dfrac{0}{0}$

$= \lim\limits_{x\to 0} \dfrac{\cos x}{3e^{3x}} = \boxed{\dfrac{1}{3}}$

(4) $\lim\limits_{x\to 0} \dfrac{\sin x - x}{x^3} = \dfrac{0}{0}$

$$= \lim_{x \to 0} \frac{\cos x - 1}{3x^2} = \frac{1-1}{0} = \frac{0}{0}$$

L'Hopital again

$$= \lim_{x \to 0} \frac{-\sin x}{6x} = \boxed{\frac{-1}{6}}$$

1

(5) $$\lim_{x \to \infty} \frac{x^3}{e^x} = \frac{\infty}{\infty}$$

$$= \lim_{x \to \infty} \frac{3x^2}{e^x} = \frac{\infty}{\infty}$$

$$= \lim_{x \to \infty} \frac{6x}{e^x} = \frac{\infty}{\infty}$$

$$= \lim_{x \to \infty} \frac{6}{e^x} = \frac{6}{\infty} = \boxed{0}$$

(6) $$\lim_{x \to \infty} \frac{e^x}{x^3} = \frac{\infty}{\infty}$$

$$= \lim_{x \to \infty} \frac{e^x}{3x^2} = \frac{\infty}{\infty}$$

$$= \lim_{x \to \infty} \frac{e^x}{6x} = \frac{\infty}{\infty}$$

$$= \lim_{x \to \infty} \frac{e^x}{6} = \boxed{\infty}$$

(7) $$\lim_{x\to 0} \frac{x\cos x-\sin x}{x\sin^2 x} = \frac{0}{0}$$

$$= \lim_{x\to 0} \frac{\cancel{\cos x} - x\sin x - \cancel{\cos x}}{\sin^{\cancel{2}} x + 2x\,\cancel{\sin x}\cos x}$$

$$= \lim_{x\to 0} \frac{-x}{\sin x + 2x\cos x} = \frac{0}{0}$$

using L'Hopital again

$$= \lim_{x\to 0} \frac{-1}{\cos x + 2\cos x - 2x\sin x}$$

$$= \boxed{\frac{-1}{3}}$$

حل آخر

$\frac{0}{0}$ $\underline{x}$ بسط ومقام

$$= \lim_{x\to 0} \frac{-1}{\underbrace{\left(\frac{\sin x}{x}\right)}_{\to 1} + 2\cos x}$$

$$= \boxed{-\frac{1}{3}}$$

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(8) $$\lim_{x\to 1} \frac{\sin \pi x}{x-1} = \frac{0}{0}$$

$$= \lim_{x\to 1} \frac{\pi \cos \pi x}{1} = \boxed{-\pi}$$

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(9) $$\lim_{x\to \infty} \frac{\ln x}{\sqrt{x}} = \frac{\infty}{\infty}$$

$$= \lim_{x\to \infty} \frac{\left(\frac{1}{x}\right)}{\left(\frac{1}{2\sqrt{x}}\right)} = \lim_{x\to\infty} \frac{2}{\sqrt{x}} = \boxed{0}$$

(10) $\lim_{x \to 1} \dfrac{\ln(\ln x)}{\ln x} = \dfrac{-\infty}{0} = \boxed{-\infty}$

(11) $\lim_{x \to 0} \dfrac{\sin(\sin x)}{\sin x} = \dfrac{0}{0}$

$= \lim_{x \to 0} \dfrac{\cos(\sin x) \cdot \cancel{\cos x}}{\cancel{\cos x}} = \boxed{1}$

(12) $\lim_{x \to 0^+} \dfrac{\sin x}{\sqrt{x}} = \dfrac{0}{0}$

$= \lim_{x \to 0^+} \dfrac{\cos x}{\frac{1}{2\sqrt{x}}} = \lim_{x \to 0^+} 2\sqrt{x}\cos x = \boxed{0}$

(13) $\lim_{x \to \infty} \left(\sqrt{x^2+1} - x\right) = \infty - \infty$

بالضرب في المرافق $\left(\sqrt{x^2+1}+x\right)$ بسط ومقام

$= \lim_{x \to \infty} \dfrac{(\cancel{x^2}+1) - \cancel{x^2}}{\sqrt{x^2+1}+x} = \dfrac{1}{\infty} = \boxed{0}$

(14) $\lim\limits_{x\to 0}\left(\frac{1}{\sqrt{x}}-\sqrt{\frac{x}{x+1}}\right) = \infty - 0 = \boxed{\infty}$

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(15) $\lim\limits_{x\to\infty} x\tan\frac{1}{x} = \infty\cdot 0$

$= \lim\limits_{x\to\infty} \frac{x}{\cot\frac{1}{x}} = \frac{\infty}{\infty}$

$= \lim\limits_{x\to\infty} \frac{1}{\cancel{-}\csc^2\frac{1}{x}\cdot\frac{\cancel{-}1}{x^2}}$

$= \lim\limits_{x\to\infty} \frac{\sin^2\frac{1}{x}}{\frac{1}{x^2}} = \frac{0}{0}$

$= \lim\limits_{x\to\infty} \left(\underbrace{\frac{\sin\frac{1}{x}}{\frac{1}{x}}}_{1}\right)^2 = \boxed{1}$

من الصور المحفوظة

حل آخر

$= \lim\limits_{x\to\infty} \frac{\tan\frac{1}{x}}{\frac{1}{x}}$

$= \boxed{1}$

من الصور المحفوظة

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(16) $\lim\limits_{x\to 0} x^2\ln x = 0\cdot -\infty$

$= \lim\limits_{x\to 0} \frac{\ln x}{1/x^2} = \frac{-\infty}{\infty}$

$= \lim\limits_{x\to 0} \frac{1/x}{-2/x^3} = \lim\limits_{x\to 0} \frac{x^2}{-2} = \boxed{0}$

(17) $\lim\limits_{x\to 0} (1-2x)^{\frac{1}{x}} = (1)^{\infty}$

let $y = \lim\limits_{x\to 0} (1-2x)^{\frac{1}{x}}$

بأخذ ln للطرفين

$\therefore \ln y = \lim\limits_{x\to 0} \frac{1}{x} \ln(1-2x)$

$= \lim\limits_{x\to 0} \frac{\ln(1-2x)}{x} = \frac{0}{0}$

$= \lim\limits_{x\to 0} \frac{\left(\frac{-2}{1-2x}\right)}{1} = \boxed{-2}$

$\therefore \boxed{y = e^{-2}}$

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(18) $\lim\limits_{x\to 0} \frac{5^x - 3^x}{x} = \frac{1-1}{0} = \frac{0}{0}$

$= \lim\limits_{x\to 0} \frac{5^x \cdot \ln 5 - 3^x \cdot \ln 3}{1} = \boxed{\ln 5 - \ln 3}$

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(19) $\lim\limits_{x \to \infty} \dfrac{3^x}{5^x} = \dfrac{\infty}{\infty}$

هنا غير مفيد تطبيق قاعدة L'Hopital لأن الناتج
دائماً سيكون $\frac{\infty}{\infty}$

$= \lim\limits_{x \to \infty} \left(\dfrac{3}{5}\right)^x = \boxed{0}$

لاحظ

$\left(\text{رقم أكبر من } 1\right)^{\infty} = \infty \qquad \left(\text{رقم أقل من } 1\right)^{\infty} = \text{Zero}$

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(20) $\lim\limits_{x \to 0} \dfrac{\cos x}{x^2} = \dfrac{1}{0} = \boxed{\infty}$

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(21) $\lim\limits_{x \to \infty} \left(1 + \dfrac{1}{x}\right)^x = (1)^{\infty}$

let $y = \lim\limits_{x \to \infty} \left(1 + \dfrac{1}{x}\right)^x$

$\ln y = \lim\limits_{x \to \infty} x \ln\left(1 + \dfrac{1}{x}\right)$

$= \lim\limits_{x \to \infty} \dfrac{\ln\left(1 + \frac{1}{x}\right)}{\frac{1}{x}} = \dfrac{0}{0}$

$= \lim\limits_{x \to \infty} \dfrac{\frac{1}{1 + \frac{1}{x}} \cdot \cancel{\left(\frac{-1}{x^2}\right)}}{\cancel{\left(-1/x^2\right)}} = 1 \quad \therefore \boxed{y = e^1}$

(22) $\lim_{x \to 0} x \cot 2x = 0 \cdot \infty$

$= \lim_{x \to 0} \frac{x}{\tan 2x} = \frac{0}{0}$

using L'Hopital

$= \lim_{x \to 0} \frac{1}{2\sec^2 2x}$

$= \boxed{\frac{1}{2}}$

حل آخر

$= \lim_{x \to 0} \frac{1}{\left(\frac{\tan 2x}{x}\right)}$

ضم الصور المحفوظة 2

$= \boxed{\frac{1}{2}}$

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(23) $\lim_{x \to 0} (\cos x)^{1/x} = (1)^{\infty}$

let $y = \lim_{x \to 0} (\cos x)^{1/x}$

$\therefore \ln y = \lim_{x \to 0} \frac{1}{x} \ln \cos x$

$= \lim_{x \to 0} \frac{\ln \cos x}{x} = \frac{0}{0}$

$= \lim_{x \to 0} \frac{\frac{1}{\cos x} \cdot - \sin x}{x} = \lim_{x \to 0} \frac{-1}{\cos x} \cdot \frac{\sin x}{x}$

$= -1$

$\therefore \boxed{y = e^{-1}}$